EXPERIMENT:-5

PROBLEM STATEMENT:-

In Linux systems, large volumes of data are often stored in text files such as logs, configuration files, and scripts. Manually searching for specific information inside these files is inefficient and error-prone. Linux provides powerful text-searching utilities that enable users to quickly locate patterns, keywords, or strings within files and command outputs.

The problem is to search for specific patterns and text within files using Linux commands such as grep, fgrep, and related options, and to understand their importance in log analysis, troubleshooting, and automation tasks.

Aim :- Deleting, Copying, Moving and Renaming files in linux. Searching for file and text using grep and related.

Procedure:-

1. Log in to the Linux system and open the terminal.
   • nanoCreate a working directory: mkdir grep_lab , cd grep_lab

```
ubuntu@seema-ubuntu:~$ mkdir grep_lab1
ubuntu@seema-ubuntu:~$ ch grep_lab1
```

2. Create a sample text file: nano sample.txt

```
Mar 30  07:48
ubuntu@seema-ubuntu: ~
  GNU nano 7.2                                   sample.txt *
linux is powerful
my name is seema
cybersecurity is my domain
```

3. Display file contents: cat file1.txt

```
ubuntu@seema-ubuntu:~$ cat sample.txt
linux is powerful
my name is seema
cybersecurity is my domain
ubuntu@seema-ubuntu:~$
```

4. Search for a word using grep: grep Linux sample.txt

grep

```
ubuntu@seema-ubuntu:~$ grep linux sample.txt
linux is powerful
ubuntu@seema-ubuntu:~$
```

5. Perform case-insensitive search: grep -i linux sample.txt

```
ubuntu@seema-ubuntu:~$ grep -i linux sample.txt
linux is powerful
ubuntu@seema-ubuntu:~$
```

6. Display line numbers with matches: grep -n Linux sample.txt

```
ubuntu@seema-ubuntu:~$ grep -n linux sample.txt
1:linux is powerful
ubuntu@seema-ubuntu:~$
```

7. Invert the match: grep -v Linux sample.txt

```
ubuntu@seema-ubuntu:~$ grep -v linux sample.txt
my name is seema
cybersecurity is my domain
ubuntu@seema-ubuntu:~$
```

8. Search using fgrep: fgrep "cybersecurity" sample.txt

```
cybersecurity is my domain
ubuntu@seema-ubuntu:~$ fgrep "cybersecurity" sample.txt
cybersecurity is my domain
ubuntu@seema-ubuntu:~$
```

9. Perform recursive search (if multiple files exist): grep -r Linux .

```
ubuntu@seema-ubuntu:~$ grep -r linux
sample.txt:linux is powerful
snap/firmware-updater/common/.cache/immodules/immodules.cache:# Created by /snap/firmwar
/lib/x86_64-linux-gnu/libgtk-3-0/gtk-query-immodules-3.0 from gtk+-3.24.38
snap/firmware-updater/common/.cache/immodules/immodules.cache:# ModulesPath = /snap/firm
usr/lib/x86_64-linux-gnu/gtk-2.0/3.0.0/linux/immodules:/snap/firmware-updater/167/gnome-
u/gtk-2.0/3.0.0/immodules:/snap/firmware-updater/167/gnome-platform/usr/lib/x86_64-linux
ap/firmware-updater/167/gnome-platform/usr/lib/x86_64-linux-gnu/gtk-2.0/immodules:/snap/
orm/usr/lib/gtk-2.0/3.0.0/linux/immodules:/snap/firmware-updater/167/gnome-platform/usr/
p/firmware-updater/167/gnome-platform/usr/lib/gtk-2.0/linux/immodules:/snap/firmware-upd
gtk-2.0/immodules:/snap/firmware-updater/167/gnome-platform/usr/lib/x86_64-linux-gnu/gtk
/firmware-updater/167/gnome-platform/usr/lib/x86_64-linux-gnu/gtk-3.0/3.0.0/immodules:/s
latform/usr/lib/x86_64-linux-gnu/gtk-3.0/linux/immodules:/snap/firmware-updater/167/gnom
gnu/gtk-3.0/immodules:/snap/firmware-updater/167/gnome-platform/usr/lib/gtk-3.0/3.0.0/li
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immodules:/snap/firmware-updater/167/gnome-platform/usr/lib/gtk-3.0/immodules:/snap/firm
usr/lib/x86_64-linux-gnu/gtk-4.0/3.0.0/linux/immodules:/snap/firmware-updater/167/gnome-
u/gtk-4.0/3.0.0/immodules:/snap/firmware-updater/167/gnome-platform/usr/lib/x86_64-linux
ap/firmware-updater/167/gnome-platform/usr/lib/x86_64-linux-gnu/gtk-4.0/immodules:/usr/li
0/linux/immodules:/usr/lib/x86_64-linux-gnu/gtk-3.0/3.0.0/immodules:/usr/lib/x86_64-linu
sr/lib/x86_64-linux-gnu/gtk-3.0/immodules
```

Expected Output

• Matching lines containing the search pattern are displayed.

• Line numbers are shown where applicable.

• Case-insensitive and inverted searches work correctly.

Result

Text searching operations were successfully performed using grep and related commands. The student gained practical experience in locating specific patterns within files, which is essential for log analysis and automation tasks.